Jet Propulsion Laboratory
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Miniaturized UHF Relay Systems for Mars Helicopters & Other Small Missions

- Mazen Shihabi (JPL) - Presenting
- UHF CMOS System Development Team: Manu Decrossas, Adrian Tang, and Zaid Towfic (JPL) Yanghyo Kim (SIT: Stevens Institute of Technology)

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The Need for Miniaturized Low SWaP* UHF Relay Systems for Small Missions (1/3)

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- Five spacecraft currently in orbit about the Red Planet make up the Mars Relay Network to transmit commands from Earth to surface missions and receive science data back from them.
- Clockwise from top left: NASA's Mars Reconnaissance Orbiter (MRO), Mars Atmospheric and Volatile EvolutioN (MAVEN), Mars Odyssey, and the European Space Agency's (ESA's) Mars Express and Trace Gas Orbiter (TGO). https://eyes.jpl.nasa.gov/apps/mrn/index.html#/mars.
- The relay link of these 5 orbiters operates in the UHF band and uses the CCSDS Proximity-1 Protocol.
- Several proposed Mars missions such as Small High Impact Energy Landing Device (SHIELD), Mars Science Helicopter (MSH), and future rovers are planning to launch small spacecrafts.
- These small spacecrafts are expected to carry several instruments to cover a larger surface of

Five Spacecraft of the Mars Relay Network

Credit: NASA/JPL-Caltech, ESA

***SWaP: Size, Weight, and Power**

SWaP UHF Relay Systems for Small Missions (2/3)

- Unfortunately, these surface spacecrafts are too small to accommodate high gain steering antennas envisioned for future MRN capabilities (e.g., X-band relay link), so a UHF relay link will likely always exist to support them (for example, NASA Mars Relay Request for Information (RFI) being drafted assumes future relay orbiters still support UHF).

- Compared to direct-to-orbit communication, small landers like Ingenuity have relied on a relay rover/lander like Perseverance to send the data back to Earth through the MRN, completely tethering them to another (larger) asset. For example, Ingenuity will lose the ability to communicate with Earth when Perseverance is out-of-range or non-functional.

Credit: NASA/JPL-Caltech, California Academy of Sciences.

The Need for Miniaturized Low SWaP UHF Relay Systems for Small Missions (3/3)

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- Although, the Ingenuity/SRH telecom link is low SWaP, but only covers ~1km of distance. The UHF relay link must communicate with orbiters on the order of ~400km+ away, making the low SWaP UHF relay link system much more difficult to accomplish than the surface links.
    - However, the UHF telecom link benefits from a free-space communication channel, as opposed to one affected by surface propagation effects.
- Hence, the need to leverage new technology and approach to create a Proximity-1 transponder that requires no more than 10W DC power consumption at a 2W RF transmit power and no more than 500g total mass (including antenna).
    - For comparison, JPL's only UHF relay/Proximity-1 transponder is currently the Electra (orbiter side: 5.1kg and requires 70W of DC power to produce 5W at RF) and Electra-Lite (rover/lander side: 3kg with 65W DC power consumption).
- Similar requirements apply to lunar landers (for example, use of CCSDS Proximity-1 protocol), but possibly at a different band (e.g., S-band).
- The support of half-duplex mode of Proximity-1 protocol by

Credit: NASA/JPL-Caltech, California Academy of Sciences.

Compact Low Energy Version of the Electra Radio (CLEVER) - 1/2

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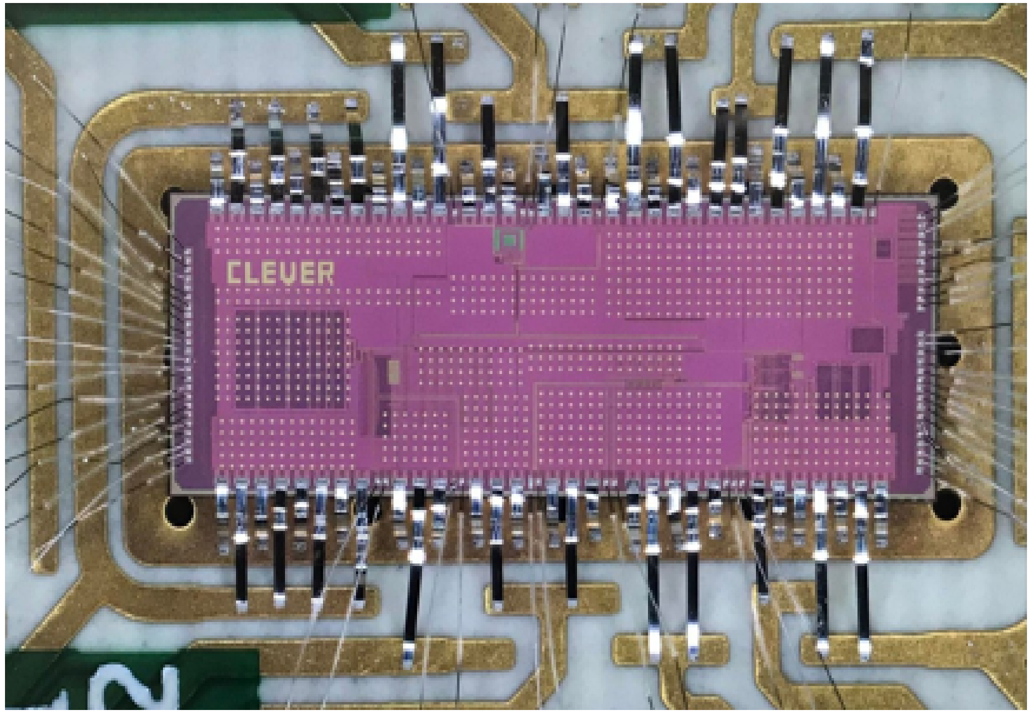
- The approach is to develop a complementary metal-oxide-semiconductor (CMOS) UHF radio-frequency integrated circuit (RFIC) to perform the down conversion, sampling, and processing part of the radio function.
- JPL, UCLA & SIT team has developed and demonstrated the Compact Low Energy Version of the Electra Radio (CLEVER) that maintains 100% compatibility with the existing Electra radio used for Mars Proximity telecom links (Mars Odyssey, MSL, MRO, M2020) and its Proximity-1 protocol while offering the mass and power reduction possible with a system-on-chip (SoC) implementation.
- The aim of CLEVER is to provide the same level of RF performance (IP3, EVM, NF) as the existing

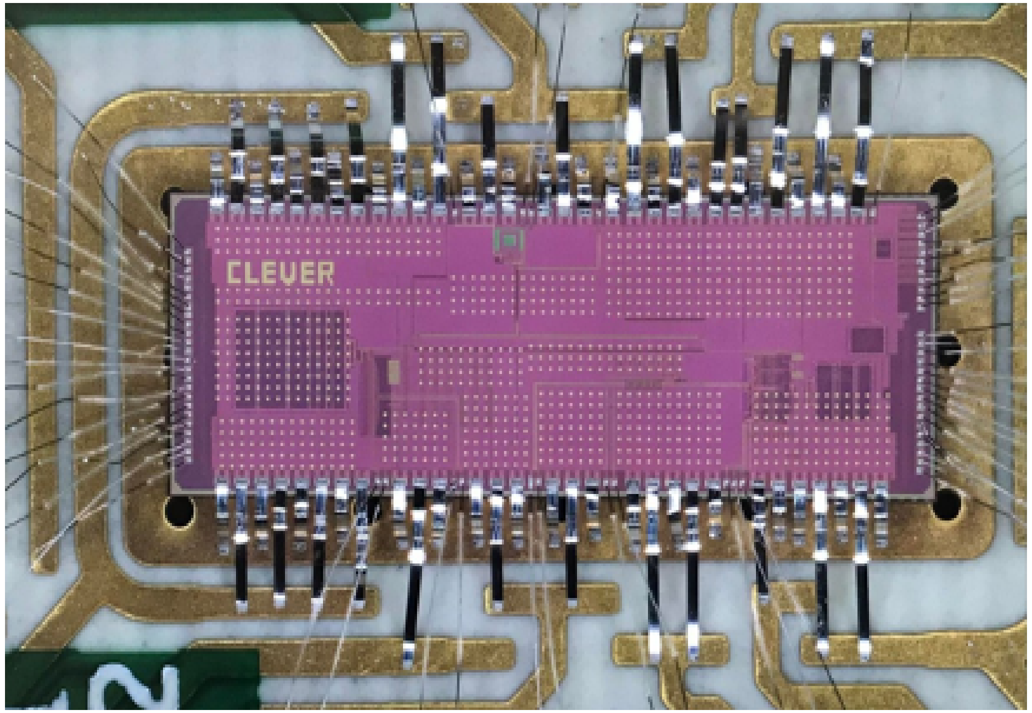


CLEVER fabricated chip

Credit: JPL, UCLA & SIT team has developed and
demonstrated the Compact Low Energy Version of the
Electra Radio (CLEVER).

Compact Low Energy Version of the Electra Radio (CLEVER) - 2/2

- The CLEVER SoC uses a radical new architecture for space radio design where none of the system is implemented as traditional analog or microwave/RF circuitry.
  - The ADC,DAC and sub-processors are implemented at such high speed (selectable from 1 to 2000 MHz) that they can directly produce and manipulate the UHF signals of the JPL Proximity-1 protocol in the digital domain.
- The nearly all-digital nature of the chip makes it relatively immune to aging effects and drastically increases its total ionizing Dose (TID) radiation robustness and immunity to large temperature ranges.
- The chip contains all the best practices for space electronics including triple-redundant registers for all configuration, packaging practices, and independent power supplies for each system block.
- The chip supports “hailing wakeup” capability where



CLEVER fabricated chip

Credit: JPL, UCLA & SIT team has developed and
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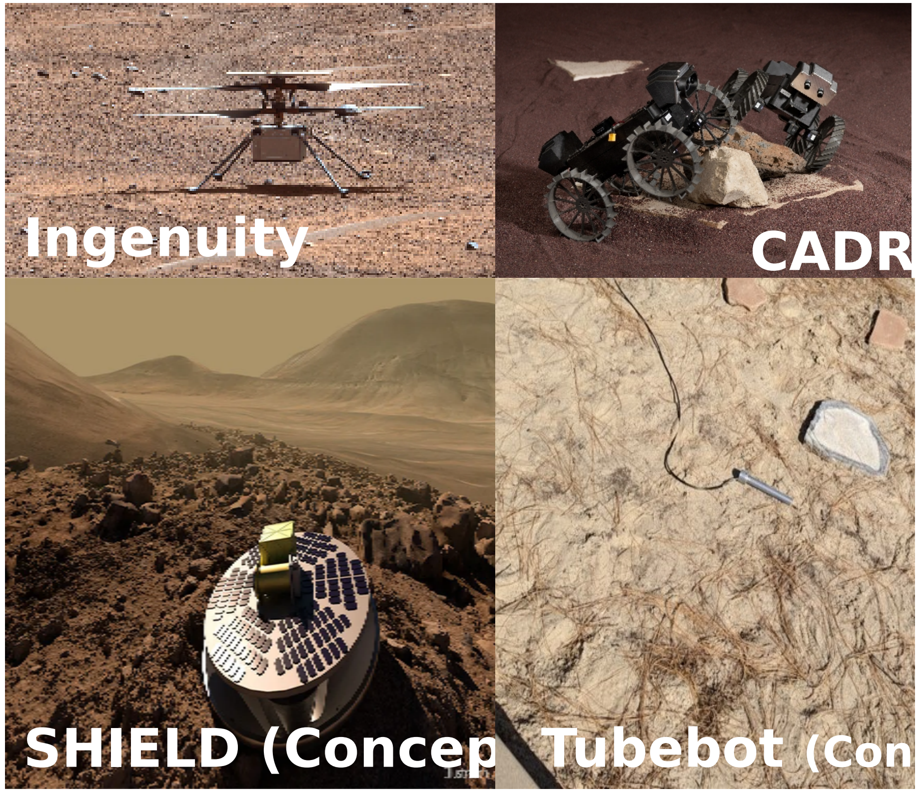
Future of Low SWaP Radios

- Current effort will bring this RFIC to TRL-4 by testing functionality to Proximity-1 waveforms and progressing TRL from there.
- Low SWaP transponders enable the smaller, cheaper, and more flexible missions.
    - Decoupling of the telecom system from larger landers/rovers allows mission ConOps to simplify radically.
- The need for low SWaP UHF and S-band transponders is driven by needs at Mars and the Moon.
    - Future effort to expand CLEVER to S-band frequencies.
- The move to Radio-Frequency Integrated Circuits has the potential to dramatically reduce size, weight, and volume requirements compared to traditional design patterns.
- Future effort is to enable half-duplex operation mode of Proximity-1 protocol for orbiters, which allows the low SWaP missions to eliminate the diplexer (a significant source of size, weight, and volume at UHF

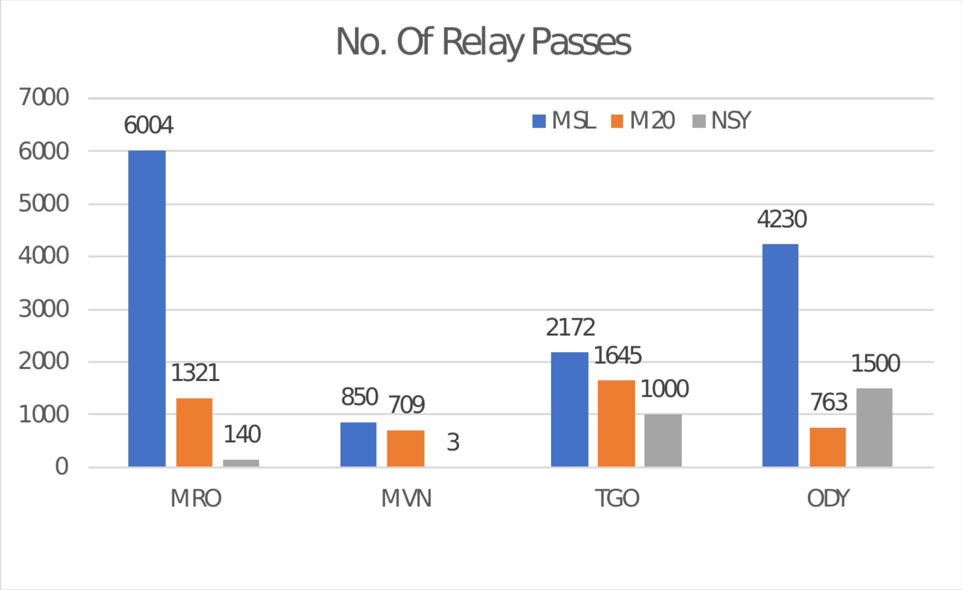


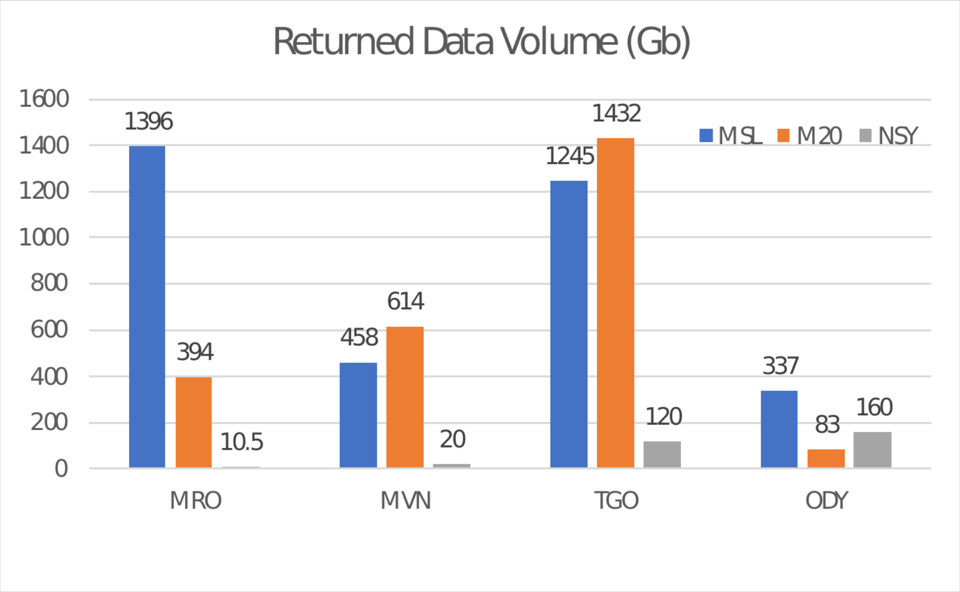
Credit: NASA/JPL-Caltech, California Academy of Sciences.

Backup

Mars Relay Network Operations Status

Orbiter	Orbit	Radio	ADR	Max Rate	LDPC
MRO	Sun sync	Electra	Yes	2 Mbps	Capable
ODY	Sun sync	CE-505	No	256 kbps	No
MVN	Elliptical	Electra	Yes	2 Mbps	Equipped
TGO	Areocentric	Electra	Yes	2 Mbps	Equipped





Total Returned DV:

- MSL: 3.436 Tbits
- M20: 2.523 Tbits
- InSight: 293.5 Gbits

Total No. Passes:

- MSL: 13,256
- M20: 4,438
- InSight: 2,660